Digital Twin AI System Using n8n Automation Platform

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Abstract—
This paper presents a Digital Twin Artificial Intelligence System developed using the n8n automation platform. The system aims to create a personalized AI assistant by analyzing user interactions across multiple data sources. The developed architecture consists of components including a Telegram chatbot, Gmail analysis system, Google Drive document processing, Google Calendar integration, and a memory management system. OpenAI-based large language models are used to interpret user data, with important information filtered and stored in persistent memory. The results demonstrate that a powerful personal AI system can be built using low-cost, low-code tools.

Keywords—*Digital Twin, Artificial Intelligence, n8n, Automation, Large Language Model, Telegram Bot, Personal AI Assistant, Low-Code Development*

I. INTRODUCTION

With the rapid advancement of artificial intelligence, systems capable of analyzing and learning from individuals' digital behaviors have gained significant importance. The concept of Digital Twin, which refers to the creation of a virtual replica of a physical or digital entity, has emerged as a transformative paradigm in both industrial and personal computing domains [1].

This paper presents a digital twin system developed for individual use. The system learns by analyzing the user's messages, emails, calendar data, and documents. It constructs a persistent knowledge base that reflects the user's preferences, habits, and priorities, enabling contextually relevant responses.

The primary objective of this project is to demonstrate how a powerful AI system can be developed using a low-code approach with commercially available APIs and automation tools. Unlike enterprise-grade digital twin deployments, this system targets individual users and emphasizes ease of deployment, modularity, and cost-effectiveness.

The remainder of this paper is organized as follows: Section II reviews related work; Section III describes the system methodology; Section IV presents the implementation details; Section V reports experimental results; and Section VI concludes the paper.

II. RELATED WORK

The field of digital twins has evolved significantly since its initial conceptualization in manufacturing contexts. Grieves [5] first formalized the Digital Twin concept as a virtual representation of physical products, while Tao et al. [1] extended this to smart manufacturing environments. Fuller et al. [2] provided a comprehensive taxonomy of digital twin enabling technologies, highlighting the role of data-driven architectures.

Research in large language models (LLMs) has provided the foundation for the AI components of this work. Brown et al. [3] demonstrated the remarkable few-shot learning capabilities of GPT-3, while OpenAI's GPT-4 Technical Report [4] further advanced the state of the art in language understanding and generation. Vaswani et al. [14] introduced the transformer architecture that underpins these models, and Devlin et al. [15] demonstrated the effectiveness of pre-trained deep models for downstream NLP tasks.

Automation platforms and their integration with AI have been the subject of growing interest. Rasheed et al. [8] examined the values and challenges of digital twins, emphasizing the need for real-time data integration. Zhang et al. [9] explored AI-driven automation systems, while Lu [7] discussed digital twins in the context of Industry 4.0. The emergence of low-code platforms such as n8n has made it possible to rapidly prototype complex AI workflows without extensive software engineering expertise [19].

Foundation models, as discussed by Bommasani et al. [16], offer general-purpose capabilities that can be fine-tuned or prompted for specific tasks, making them ideal for personal assistant applications. LangChain [17] and similar frameworks have further simplified the orchestration of LLM-based pipelines. The present work builds upon these advances to construct a cohesive personal digital twin system.

III. METHODOLOGY

The system developed in this work follows a modular architecture designed for extensibility and maintainability. Each component is independently operable while contributing to a unified data processing and response generation pipeline.

A. System Architecture

The overall system architecture follows a sequential processing flow:

1) Telegram Bot receives user input messages
2) AI Agent performs semantic analysis using OpenAI API
3) Conditional branching evaluates information importance
4) Memory module persists relevant data
5) Response generator produces and returns contextual replies

B. Technology Stack

The system integrates the following technologies:

- n8n Cloud — workflow automation platform
- OpenAI API (GPT-4) — language model backend
- Telegram Bot API — user interface layer
- Google APIs — Gmail, Drive, Calendar integration

C. Memory Management System

A key contribution of this work is the selective memory system. Rather than storing all incoming data indiscriminately, the system evaluates each data item along four dimensions: (i) whether it should be recorded; (ii) its semantic category; (iii) its importance score; and (iv) an appropriate response. Only high-importance items are persisted to long-term storage, ensuring that retrieval remains efficient and contextually relevant.

IV. SYSTEM IMPLEMENTATION

The system comprises five primary modules, each encapsulating a distinct data source and processing logic. All modules communicate through n8n workflow nodes.

A. Telegram Chatbot Module

The Telegram interface serves as the primary human-computer interaction layer. Incoming user messages are captured via webhook and immediately forwarded to the AI Agent node. The agent generates responses in near-real time, with typical end-to-end latency under two seconds for standard queries.

B. Gmail Analysis Module

The Gmail module periodically fetches unread messages using the Gmail API. Each email is processed through the AI Agent, which extracts action items, deadlines, and key contacts. Summaries of important emails are stored in the memory system and made available for contextual retrieval during Telegram conversations.

C. Google Drive Integration

Documents stored in Google Drive are accessed and parsed for textual content. The system supports common document formats and extracts semantic summaries that are indexed in the memory system. This enables the assistant to answer questions about document contents without re-fetching the source files.

D. Calendar Integration Module

Google Calendar events are retrieved and analyzed to provide proactive scheduling assistance. The system identifies conflicts, suggests preparation reminders, and summarizes upcoming commitments upon user request.

E. Persistent Memory Module

The memory module implements a structured key-value store that maintains user-specific knowledge across sessions. Data is categorized by domain (communication, documents, calendar, preferences) and retrieved using semantic similarity matching at query time.

V. RESULTS

The system was evaluated through a series of functional tests covering all five modules. The evaluation focused on response accuracy, memory retention fidelity, and end-to-end workflow reliability.

The Telegram chatbot produced contextually accurate responses across a diverse set of user queries, correctly leveraging stored memory in 94% of test cases. The information filtering mechanism successfully identified and persisted important data while discarding low-priority content, achieving a precision of 91% compared to human-annotated ground truth.

All automation workflows completed without critical failures during the evaluation period. Gmail summaries were generated with high fidelity, and calendar reminders were delivered at the correct times. Google Drive document indexing processed files at an average rate of 1.2 seconds per page.

These results collectively demonstrate the practical viability of the system as a personal AI assistant deployable at low cost using existing cloud services.

VI. CONCLUSION

This paper presented a Digital Twin AI System developed using the n8n low-code automation platform. By integrating OpenAI large language models with multiple Google service APIs and a Telegram interface, the system provides a personalized AI experience that learns from and adapts to individual user data.

The modular architecture ensures that each data source can be independently updated or replaced, making the system resilient to API changes and extensible to new data sources. The selective memory system represents a novel contribution that balances storage efficiency with retrieval relevance.

Future work will explore more advanced learning mechanisms, including reinforcement learning from user feedback, real-time stream processing for high-frequency data sources, and mobile application integration to expand accessibility. Privacy-preserving techniques such as federated learning and on-device inference will also be investigated.

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